Tomorrow's tech
44 The future of motor vehicles seems to be quite promising. We take a look at some of the upcoming motoring innovations.

A hundred million
...solar pictures were taken by a NASA space telescope
http://dgit.in/100millpics

Commuting to space

... soon a reality

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Mankind has always had a deep fascination for the stars and planets. A strong belief in astrology and the acceptance that planetary changes affect one's fate to the curiosity of finding out what lies beyond the stratosphere - space has always been the final frontier. But will travel into that great infinity that is space and witness the wonders of the universe? Well, soon! Looking at the bustling activity in the space corridors of power, commercial space travel may soon be the real deal. Virgin Galactic has already announced that they plan to start a round trip service to the outer space. What's more, they are in the process of building the first ever spaceport for commercial space transport. And, with a fleet of SpaceShipTwo reusable space vehicles on standby, outer space may soon become the new travel destination of choice.

XCOR's Lynx Mark I is scheduled for flight testing this year

LONG WAY TO SPACE

However, soon may not really be soon enough. Despite breakthroughs in the field, many technical and logistical challenges still need to be ironed out. For instance, the biggest challenge surrounding commercial space travel is the lack of reusable space vehicles. The chequered past of reusable spacecraft is an indication that there's still some time to get the space act right. The 1990s witnessed a slew of companies, rushing in to get a piece of the space pie. Many startups appeared to create reusable launch vehicles (RLVs) but the stiff standards and competition coupled with poor management and flawed design saw many fall by the wayside or go into a limbo. The failure of Nasa's X-33 and X-34 missions also didn't help the cause. After a few years of hesitation, the commercial space travel plan in back on track with a lot more research and attempts are creating reusable space vehicles, to reduce the cost of operations in the long run.

Companies like XCOR Aerospace and The Spaceship Company are creating cutting edge reusable spacecraft models and some of it's currently being tested. All eyes are on the Orion Multipurpose Crew Vehicle, the first reusable space vehicle by NASA after their Spaceshuttle was retired in 2011. The Orion has an impressive spec sheet and builds on the Spaceshuttle design and technology. The "green" spacecraft is built to carry crew to space, provide emergency abort capabilities, sustain the crew and provide a safe return to Earth. The first flight test was done on 5 December 2014, where the unmanned Orion was launched atop the Delta IV Heavy rocket from Cape Canaveral's Air Force Station's Space Launch Complex 37. The spacecraft was loaded with almost 1,200 sensors and it completed a successful 4.5 hour flight. In the future, Orion will be launched on Nasa's new heavy-lift rocket, the Space Launch System.

X-RATED

Yet another spacecraft that is the subject of much conversation is the XCOR Lynx, a suborbital horizontal-takeoff, horizontal-landing (HTHL) spaceplane that is currently being developed by XCOR Aerospace, a California-based company. XCOR says that the Lynx can reportedly fly four or more times in a day and can even deliver payloads in space. The Lynx